

Investigating the Impact of Multimodal Feedback on User-Perceived Latency and Immersion with LLM-Powered Embodied Conversational Agents in Virtual Reality

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ABSTRACT

Our research investigates the impact of latency on presence and immersion in virtual reality (VR) environments, focusing on interactions with LLM-powered Embodied Conversational Agents (ECAs). We explore the effectiveness of multimodal feedback strategies—including filled pauses, nonverbal turn-taking behaviours, and visual feedback—in mitigating perceived latency. Eighteen participants were subjected to both a baseline condition, without feedback interventions, and a feedback-enhanced condition. Our findings indicate that the feedback condition significantly improved the sense of presence and immersion. We also found that perceived response time and users’ impressions of the agents improved, thereby increasing willingness for future interactions. Additionally, chatbot experience positively correlated with agent likeability, whereas VR experience showed no significant correlation. These results highlight the effectiveness of feedback modalities in enhancing spatial presence and overall immersion, despite latency issues in VR interactions with LLM-powered agents.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

Embodied Conversational Agents, Virtual Reality, Presence and Immersion, Latency

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1 INTRODUCTION

Embodied conversational agents are “computer-generated cartoon-like characters that demonstrate many of the same properties as

humans in face-to-face conversation, including the ability to produce and respond to verbal and nonverbal communication” [5].

Recent advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP) have enabled the development of ECAs that are more lifelike and intelligent than their earlier counterparts., capable of engaging in meaningful conversations and exhibiting behaviours that mimic real human interactions [24]. In particular, the integration of large language models (LLMs) into the design of ECAs has marked a significant leap in the field, enabling a level of conversational intelligence and responsiveness previously unattainable with traditional rule-based or statistical approaches [6]. The capacity of LLMs to generate dynamic, context-aware dialogue has not only enhanced the user experience but also broadened the applications of these agents in sectors ranging from customer service to therapeutic support [17, 18].

Despite these advancements, the deployment of LLM-powered agents faces a critical challenge: the latency introduced by reliance on cloud-based operations for core functionalities such as transcribing user speech through Automatic Speech Recognition (ASR), comprehending it using Natural Language Understanding (NLU), as well as creating text responses to feed back to the user via natural language generation (NLG) and using Text-to-Speech (TTS) services for synthesising audio [7, 37]. Cloud-based operations come with the benefit of low local processing performance for users, as well as greater external provider accuracy in language processing tasks; the trade-off is increased variability in system response time (SRT)—the elapsed time between a user’s query and an agent’s response.

Latency significantly impacts the immediacy and fluidity of interactions between users and ECAs [20]. Hence, timely and coherent responses are crucial for maintaining user engagement and satisfaction. When a user interacts with an agent, these sequential system components may experience unpredictable latency times which add up, ultimately resulting in response delays for the user. This challenge is particularly relevant to the domain of virtual reality (VR), where the smoothness of real-time interactions is paramount to enhancing users’ experiences of immersion and presence.

In recent years, access to VR technology has grown significantly in the consumer market, making immersive experiences more attainable and appealing across various domains. As VR continues to gain traction, its potential applications in fields such as training [8], education [26], and therapy [27] are expanding rapidly. Within this evolving landscape, the role of ECAs becomes critically important, as they may serve as the primary interface through which users interact with the virtual world.



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The benefit of virtual reality is seen in its ability to create immersive experiences that significantly enhance the user's sense of presence—the feeling of being "there" in a digitally created environment [30]. Platforms which can deliver a vivid illusion of reality to the users' senses are considered more immersive if they can display sensory modalities while maintaining fidelity in relation to their equivalent real-world sensory counterparts [36]. Hence, elements such as audio and visuals affect the degree of immersion provided by a system [11].

However, presence is a dimensional construct which measures how present a user feels (subjectively) within a virtual environment. It comprises three different categories [21]: Spatial Presence—the degree to which a person feels like they're in the virtual environment instead of the real world; Self Presence—the feeling that one's virtual representation aligns closely with the actual self; and Social Presence—the feeling of being together with others, characterised by how easily one can sense another's thoughts, intentions, and emotions.

Increased social presence can help to predict positive communication outcomes [30]. Considering that social presence of virtual characters is greater in VR environments compared to traditional flatscreen systems [15], it's relevant to consider the impact of latency on user experience in VR environments integrating semantically-sophisticated LLM-powered agents because delays in agent responses have the potential to disrupt user presence, immersion, and overall interaction flow.

Previous research has highlighted the adverse effects of latency on users' perceptions of agents, noting a marked decrease in user satisfaction as response times exceed user expectations [28, 42]. Consequently, strategies such as the usage of filled pauses and nonverbal turn-taking behaviours have been explored as ways of maintaining user engagement during delays [4, 19]. However, the application of these strategies and their effectiveness in mitigating latency perceptions within the context of LLM-powered agents within VR environments remains unexplored. Thus, understanding and optimising social presence alongside immersion and latency management is essential for creating compelling and engaging VR applications.

1.1 Research Contribution

This research addresses a critical yet underexplored aspect of VR experiences: the impact of latency in LLM-powered agents on user immersion. It investigates how existing strategies to reduce perceived latency perception—namely filled pauses and nonverbal turn-taking behaviours—can mitigate the negative effects of latency on users' sense of immersion in VR and offers insights into the unique challenges and opportunities presented when interacting with agents within virtual environments. Furthermore, this research contributes to the literature by investigating how visual feedback indicators—graphical indicators that signify ongoing processes, used in many systems though not yet studied alongside filled-pauses and nonverbal turn-taking behaviours, might help to mitigate users' sense of perceived latency and immersion.

This research advances the dialogue on improving ECA responsiveness and user experience, enhancing VR's effectiveness for

human-agent interaction. Hence, we address the following research questions:

RQ1 To what extent does latency in LLM-powered agents' responses impact individuals' sense of immersion and presence in VR?

RQ2 How does the integration of filled pauses and nonverbal turn-taking behaviours with visual feedback indicators affect perceived latency when interacting with such agents in VR?

RQ3 To what extent does the combined feedback modalities affect participants' impressions, likeability, and willingness to interact again with the agent?

RQ4 Does level of experience using chatbots or VR affect individuals' impressions of such agents when delays occur?

2 RELATED WORK

2.1 Delays in Embodied Conversational Agents and Impact on User Perception

In dialogues between humans, pauses and silences are natural phenomena [10, 16, 38]. When applied to conversational systems they have been shown to augment the perceived naturalness of interactions [2, 13, 33]. However, excessive delays tend to pose a noticeably negative impact on users' experience and perception of agents. For instance, Shiwa et al. [34] found that users preferred a response time of approximately one second when interacting with communication robots. And although humans are able to adapt and tolerate these longer response times, a quicker reply is usually preferred [35].

Various studies have highlighted that response time delays can hinder user experience and their evaluation of not just the agent but also the entire interaction. Yang et al. [20] noted that delays in a remote-controlled agent's response time increased user frustration and reduced satisfaction; similarly in Andreea et al.'s study, which investigated the deployment of an agent in a public space [11], a key factor which influenced participants' perception of the agent and interaction quality was the agent's rapidity: being more prompt resulted in less user frustration and increased tolerance of its faults.

2.2 Using Filled Pauses to Reduce Perceived Latency

Previous literature has considered the usage of filled pauses (for instance, 'uh' or 'let me think') as a delay mitigation strategy [4, 12, 19, 23, 41]; these sorts of fillers are natural in human conversation and are used to serve various functions such as conveying agreement, reducing awkward silences or controlling turns in a conversation [25, 31]. Wigdor et al. [41], for example, showed that filled pauses when combined with gestures such as nodding can help in mitigating negative perceptions of response delays on user experience and agent impression. Halim et al. [4] investigated two types of filled pauses: generic utterances which simply asked users to wait and contextualised fillers which expressed a degree of semantic knowledge of users' inputs with some of it included in the agent's response. Their results showed contextualised fillers positively impact participants' impressions of an agent's response time but have no effect on its overall likeability. As well, Gambino et al. [23] found that, given equal response times, users overall prefer a

system which integrates filled pauses and acknowledgements of users' queries over a system that requests users to silently wait.

2.3 Using Non-Verbal Cues to Reduce Perceived Latency

Non-verbal cues such as gestures, eye gaze, facial expressions, and posture, are integral to enhancing real-world communication and rapport building [40]. Studies have shown that such non-verbal turn-taking behaviours aid in facilitating understanding and emotional connection more effectively than verbal communication alone. For instance, it has been reported that visual cues significantly boost comprehension of spoken content, especially in noisy environments [25]. Further studies show that naturalness, emotional connectivity and fluidity of interaction are improved when agents express non-verbal cues such as nodding and glancing [14], as well as mimic users' facial expressions and intonations [1]. However, a disconnect between speech and gestures can negatively impact interlocutors' perceptions [9], highlighting the significance of alignment between verbal and non-verbal communication. Kum et al. [19] investigated the impact of matching conversation types with nonverbal turn-taking behaviours and compared them with filled pauses, revealing that nonverbal turn-taking behaviours effectively reduce user-perceived latency and influence users' willingness, impressions, perceived competence, and discomfort during interactions with ECAs.

2.4 Summary

To summarise, minimising response time delays and effectively utilising filled pauses and nonverbal turn-taking cues can enhance user satisfaction and perceived naturalness of interactions with ECAs. This research extends on prior work by examining the effect of latency in LLM-powered agents within VR environments. It explores how filled pauses and nonverbal turn-taking behaviours can reduce perceived latency and its adverse effects on agent impression, presence and immersion. By also considering the role of a visual feedback indicator—previously unexplored in this context—it offers new insights into improving users' experiences with such agents in VR.

Our hypotheses were:

H1: The combined feedback modalities will have a positive effect on participants' sense of presence and immersion.

H2: The combined feedback modalities will improve participants' perception of the agent's response time.

H3: The combined feedback modalities will generally improve participants' impression, likeability, and willingness to interact again with the agent.

H4: Participants with higher chatbot and VR usage would show greater agent likeability and be more willing to interact with the agent again in the future.

3 METHODOLOGY

This study aimed to explore the effectiveness of a multimodal feedback strategy in mitigating negative perceptions of response time delays during interactions with ECAs. Participants engaged in a conversation with an agent, structured as a role-play scenario where they discussed challenges related to managing heavy workloads

from school or work. The conversation was designed to mimic a supportive dialogue, with the agent providing advice and encouragement.

To address the issue of perceived delays, the experiment incorporated two types of conversational fillers—filled pauses and nonverbal gestural cues—along with a visual feedback icon indicating the agent's processing state. This setup aimed to create an immersive interaction environment to help investigate how combining filled pauses and nonverbal turn-taking behaviours, complemented by visual feedback, can alleviate users' sensitivity to response latencies in a context mirroring real-world engagements. Subsequent analysis focused on assessing whether this integrated feedback approach enhanced users' sense of presence and immersion, altered perceptions of latency, and impacted overall impressions of the agent, providing valuable insights into the design of more responsive and engaging agents.

3.1 Design

The study used a within-subjects design in which participants conversed with an agent named Alex within a VR environment. This included two conditions: a baseline condition with none of the feedback modalities present and another with all feedback modalities present (filled pauses and nonverbal turn-taking behaviours, and a visual feedback indicator). During the conversation, participants were tasked with playing the role of either a student or professional that was struggling to keep up with the demands of their studies or work. Meanwhile, the role of Alex was to provide support and advice to participants on how to manage their workload. The VR environment used was a cosy cabin, suitable for a conversational setting. The agent's appearance was designed using the Ready Player Me SDK ¹.

OpenAI's Whisper and TTS APIs were used to transcribe participants' audio and send a TTS response back to them. For the lip-syncing, the Oculus Lipsync Unity package was used ². The agent's verbal responses were generated using OpenAI's GPT 3.5-Turbo language model—a decoder-only transformer model that uses autoregressive generation given a context, which can produce very natural speech in response to users' queries. Open AI's services were used as they provided the most streamlined integration for connecting together the various modules; the chosen model presented the least amount of friction in terms of integration in Unity while also being the most cost-effective.

To foster realistic dialogue, participants were not restricted to a predetermined script but instead were encouraged to converse naturally with the agent. The general structure and flow of conversation was: greeting (start), mentioning of the problem and main dialogue (middle), and goodbye (end). This role-play scenario was selected for its relatability to real-world experiences, ensuring it was general yet not so sensitive as to resemble professional therapeutic sessions. Moreover, through this design it would be possible to obtain more practical insights into how individuals would genuinely interact with such a system. An excerpt of a conversation with the agent is shown in Figure 1.

¹<https://docs.readyplayer.me/ready-player-me/integration-guides/unity>

²<https://developer.oculus.com/downloads/package/oculus-lipsync-unity/>

In selecting this setting, we considered its potential relevance for future applications of ECAs and its resonance with overburdened students, who comprised our participant pool; but our general research questions on latency stand for any conversational domain and use case of ECAs.

Participant: Hello.
 Alex: Hello. How are you feeling today?
 Participant: Participant: I am looking for help with managing my work world at university.
 Alex: I'm glad you reached out for help. Managing a heavy workload at university can be challenging. How would you describe your current approach to managing your tasks and responsibilities?
 Participant: At the moment, I just try and do as much as I can in the time that I have but it's not working out very well for me. I'm quite stressed.
 Alex: It sounds like you might be feeling overwhelmed with your workload and the current way of approaching tasks is causing stress. Have you considered breaking down your tasks into smaller, more manageable parts to work on gradually?
 Participant: Yes, I have tried breaking down my tasks but I struggle with prioritising which ones I should do first and which ones are most important.
 Alex: It's understandable to feel unsure about task prioritization, especially with a heavy workload. Have you thought about creating a list or schedule that outlines all your tasks and deadlines, so you can visually see what needs to be done first?
 Participant: It's a good idea. I have tried laying out a schedule for myself, but I often struggle to stick to that schedule and then get things done.
 Alex: Consistency is key when it comes to sticking to a schedule. Perhaps setting smaller, achievable goals within that schedule could help you stay on track. How do you feel about incorporating short breaks between tasks to avoid burnout and stay focused?
 Participant: Something that I could try. The thing is, when I take a break sometimes, I find it hard to start working again.
 Alex: Taking breaks is essential for productivity, but you're right that getting back into work mode can be challenging. Have you considered incorporating different types of breaks, like short physical exercises or mindfulness practices, to help refresh your mind and body for the next task?
 Participant: That sounds like a good idea. Could you give me a couple of practical examples of short breaks that I could take?
 Alex: Absolutely, you could try a quick stretch break, a short walk outside to get some fresh air, or even a five-minute mindfulness exercise to relax your mind. How do you feel about incorporating these short breaks into your schedule to see if they help you stay refreshed and focused?
 Participant: It was a really good idea. Do you have any stretching routines that you could share with me? I often get neck and back pain when I'm studying for long periods of time.
 Alex: I'm glad you found the idea helpful. For neck and back pain relief during study sessions, you could try simple stretches like neck rolls, shoulder stretches, and seated twists. Would you like me to guide you through these stretches now, or would you prefer some resources or videos to refer to later?
 Participant: Yeah, could you guide me through these exercises now?
 Alex: Of course! Let's start with some gentle neck rolls. Slowly tilt your head to one side, bringing your ear towards your shoulder, and then gently roll your head to the other side. Repeat this movement a few times on each side to release tension in your neck. How does that feel so far?
 Participant: That was really good, thanks for the help.
 Alex: You're welcome! If you need anything else, feel free to ask.
 Participant: Bye for now.
 Alex: Bye for now! Have a great day!

Figure 1: Excerpt from a conversation with the agent

By using an LLM to power an agent's dialogue, this introduces the potential risk of something undesirable being said. To mitigate this risk as much as possible, a system prompt was provided to the agent, providing it with context about the roleplay scenario, its role, and how it should behave throughout the interaction. An example of the prompt is displayed below:

"Your name is Alex. You're part of a role play scenario. You're in a cosy cabin home environment with another person. A conversation takes place between the two of you and they mention that they're having trouble managing their heavy workload so try to provide them with support. If the person you're speaking with says more, then you should too and vice versa. Write your responses similar to what natural spoken speech sounds like and do not mention you're an AI."

Both the participant and agent were seated during the interaction. Mocap Fusion VR was used³ to create custom motion capture for the nonverbal turn-taking behaviours as there were no freely accessible animations online which suited this study. To create the motion capture, the Meta Quest 2 was used (see Figure 2), alongside x3 VIVE Tracker 3.0s (for seating the agent) and two Valve Index controllers (used instead of the Meta Quest 2 controllers as they provide a wider range of tracking motion). Two filler gestures were created (see Figure 3): one where the agent looks down, tilting their head slightly as if thinking, and another where the agent places their hand behind their head when posed a question. The chosen gestural fillers aimed to convey that the agent is preparing its response and reflecting on the user's query. Once created, the nonverbal turn-taking behaviours were imported into Unity, which was the main tool used to develop the system. Additionally, a visual feedback indicator was implemented, shown as a grey spinner icon located to the right of the agent's head in Figure 3. This feature was designed to signal processing or thinking.

The filled pauses used in the condition with the feedback modalities present were uncontextualised filled pauses; this included generic utterances such as "hm" and "let me think".

³<https://www.mocapfusion.com/>



Figure 2: Mocap Fusion VR and motion capture equipment

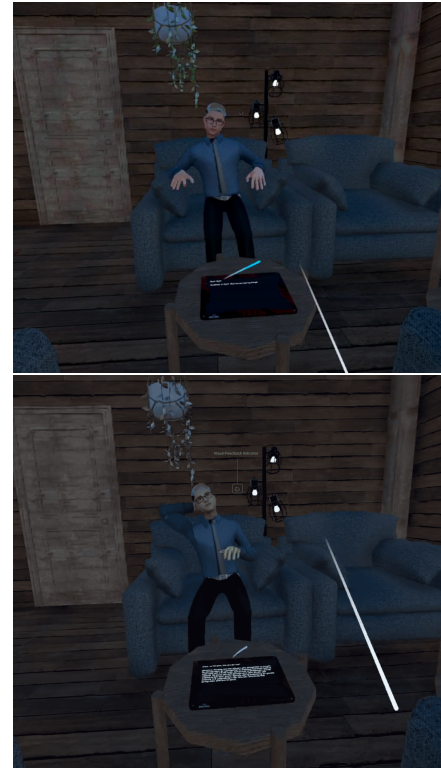


Figure 3: Gestural animations and visual feedback indicator

3.2 Participant Recruitment

Eighteen participants of varying cultural and social backgrounds were recruited via email, word of mouth and advertising in University Teams channels (Age mean=22.1, SD=2.8, 14 male, 3 female, 1 non-binary). To participate, participants were required to provide informed consent. Moreover, participants with serious or professionally diagnosed mental health issues were excluded from participating to avoid potential psychological triggers from the LLM's unpredictable output. To minimise order effects, participants were counterbalanced and assigned to either the condition with the feedback modalities absent or the condition with them present as their initial exposure. Ethical approval for this study was secured via the appropriate university channels.

3.3 Procedure

Participants were provided with an information sheet to learn about the purpose of the study before opting to participate. After providing informed consent, participants were asked to fill in a pre-survey questionnaire which asked about demographics, VR and chatbot usage; additionally, the Negative Attitudes Towards Robots (NARS) questionnaire [29] was used to measure participants' perceptions of ECAs. The distribution of participants' VR and chatbot experiences can be seen in Figure 4.

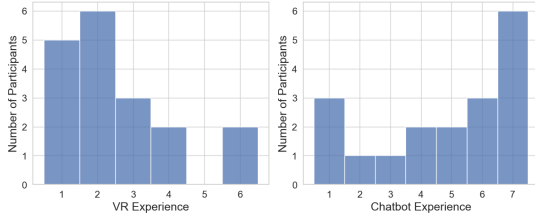


Figure 4: Participants' VR and Chatbot Experiences (1=never, 4=monthly, 7=almost daily)

After completing the pre-questionnaire, participants were helped with setting up the VR device and a short tutorial was given explaining how to interact with the agent in VR. Participants were additionally made aware that the agent was an AI. To replicate the virtual setting, which was designed to include only the user and the agent, participants were left alone in the experiment room to interact without external distractions that could influence their engagement with the agent.

To start the interaction, participants greeted the agent before addressing the study or work-related issues they were encountering. The number of turns in the dialogue was not restricted so as to allow participants more flexibility to convey themselves naturally. With respect to time, each session lasted for approximately seven minutes to allow participants enough opportunity to interact with the agent in a meaningful way to form an impression of it and their experience.

After a participant finished their turn by pressing on a button mapped to the VR controller, a 2.5s wait time was enforced for the feedback modalities condition before the filled pause was played. This interval was chosen due to being consistent with a previous study [4]. During piloting, participants (excluded from the main study) reported that playing a filled pause at each turn felt somewhat irritating and less natural; since the agent's response from the LLM sometimes arrived before 2.5s, we chose to skip playing the filled pause if the delay time wasn't too long ($< 2.5s$). As well, a delay of 1.5s was added between the filled pause and the TTS response since playing the TTS audio immediately after felt abrupt and unnatural.

After each condition, participants were asked to fill out a post-questionnaire which used questions from the Igroup Presence Questionnaire (IPQ) [32] and The Temple Presence Inventory (TPI) [22] to measure the sense of presence and immersion experienced in the virtual environment. Questions from the Godspeed Questionnaire Series [3] were also used to evaluate participants' impressions of the agent across both conditions. Additional questions inquired about

participants' opinions on the agent's response time, their willingness to interact again, its likability, and how freely they felt they could express themselves to the agent. After completing both conditions, participants were asked to identify the feedback modalities used to ensure they were aware of the specific modalities used, and then rank them by preference. Finally, a semi-structured interview was conducted to provide qualitative insights on the effectiveness of the feedback modalities and an understanding of participants' rankings.

3.4 Limitations

Limitations of this study include its sample size and choosing to investigate all feedback modalities together rather than each one separately. By focusing on the two most extreme combinations of conditions, it was possible to identify significant impacts without the complexity of multiple variable interactions. Ideally, each modality would be examined separately, which would require more time from participants.

In our study, we used a standalone VR headset due to resource constraints. This choice, while practical, imposed graphical fidelity limitations, potentially compromising the immersive quality of the virtual environment—a critical factor for immersion [11]. Our inability to incorporate subtle gestures like gaze and facial expressions, significant for conveying non-verbal behaviours [40], further constrained the depth of interaction as we lacked the skillset to implement these at the time. Instead, we relied on random eye movement and basic lip-syncing. The creation of custom motion capture animations was limited; only two animations were captured due to inexperience and technical challenges, such as unnatural clipping issues when attempting gestures like placing a hand on the chin caused by limitations in the agent's physics system.

The implementation of a single character design might have affected users' impressions, considering individuals may perceive appearances differently [39]. Convenience sampling was used as this was most practical given the lack of financial incentive to participate; resultingly, the majority of participants were young students or graduates. Additionally, participants were aware that their interactions within the headset would be recorded. A tablet displaying dialogue transcriptions, was placed on a small table between the participant and agent in a diegetic manner, indicating recording of interactions; this awareness may have influenced participants' willingness to express themselves fully, affecting our measures of openness to express.

For filled pauses, length and tone act as crucial non-lexical cues for expressing emotions in dialogues [19]. To align with the TTS voice model (OpenAI's Alloy) used in our study, we opted for pre-recorded filled pauses in the same voice. This choice constrained our ability to control intonation, tone, and style, thus limiting the range of emotions that could be conveyed through these non-lexical cues. While we tried to ensure that the fillers varied in sound to avoid monotony by capturing different recordings of the same words, techniques such as round-robin sampling and randomised frequency filtering hadn't been used. Moreover, the necessity for pre-recording TTS responses inhibited the integration of contextualised filled pauses, which leverage semantic understanding of users' queries and incorporate elements of it [4].

4 RESULTS

4.1 Effect of Latency on Individuals' Sense of Immersion and Presence

Table 1 compares participant scores for level of immersion and presence experienced between the two conditions. Statistical analysis was performed using a repeated-measures t-test.

Table 1: Means (Std.); Comparison of TPI and IPQ Scores Reflecting Social Presence, Engagement, Social Realism, General Presence, Spatial Presence, Involvement, and Experienced Realism Based on 7-Point Likert Scale

Factor	1: Base Condition	2: Modals Condition	p-value
Social Presence	4.74 (1.19)	5.48 (0.83)	0.03
Engagement	4.67 (1.11)	5.18 (0.97)	0.05
Social Realism	4.85 (1.36)	5.44 (1.15)	0.02
General Presence	4.44 (1.58)	5.33 (1.14)	0.04
Spatial Presence	4.81 (1.31)	5.28 (1.14)	0.04
Involvement	3.94 (0.41)	3.82 (0.78)	0.46
Experienced Realism	4.36 (1.14)	4.58 (1.03)	0.33

There was a significant improvement in perceived social presence in the feedback modalities condition ($M = 5.48$, $SD = 0.83$) compared to the base condition ($M = 4.74$, $SD = 1.19$, $p = 0.03$), indicating that participants seemed to feel a greater sense of co-presence with the agent inside of the virtual environment.

Similarly, engagement scores were noticeably higher in the feedback modalities condition ($M = 5.18$, $SD = 0.97$) compared to the base condition ($M = 4.67$, $SD = 1.11$, $p = 0.05$), implying that the multimodal feedback improved mental immersion.

Participants also reported a higher degree of social realism felt in the feedback modalities ($M = 5.44$, $SD = 1.15$, $p = 0.02$) compared to the base condition ($M = 4.85$, $SD = 1.36$) highlighting an increase in the believability of the interaction's realism.

Scores for general and spatial presence were likewise higher in the feedback modalities condition (general presence: $M = 5.33$, $SD = 1.14$, $p = 0.04$; spatial presence: $M = 5.28$, $SD = 1.14$, $p = 0.04$), showing a higher degree of presence was felt within the virtual environment.

As for involvement and experienced realism, neither showed a significant change: involvement showed a slight decrease in the feedback modalities condition ($M = 3.82$, $SD = 0.78$) compared to the base condition ($M = 3.94$, $SD = 0.41$) while experienced realism observed a minor increase from the base condition ($M = 4.36$, $SD = 1.14$) to the feedback modalities condition ($M = 4.58$, $SD = 1.03$).

4.2 Effect of Combined Feedback Modalities on Perceived Agent Response Time

There was a significant increase in the perceived appropriateness of the agent's response time in the feedback modalities condition

($M = 3.50$, $SD = 0.86$, $p = 0.02$), compared to the base condition ($M = 2.72$, $SD = 1.27$). The results are shown in Table 2 (1=strongly disagree, 5=strongly agree).

Table 2: Perceived Appropriateness of Agent Response Time

1: Base Condition	2: Modals Condition	p-value
2.72 (1.27)	3.50 (0.86)	0.02

4.3 Effect of Combined Feedback Modalities on Perceived Impression of the Agent

Table 3 summarises the results obtained from the Godspeed Questionnaire for both conditions.

Table 3: Godspeed Results for Anthropomorphism, Animacy, Likeability, Perceived Intelligence, and Safety

Factor	1: Base Condition	2: Modals Condition	p-value
Anthropomorphism	2.58 (0.65)	3.06 (0.71)	0.01
Animacy	3.09 (0.59)	3.58 (0.68)	0.01
Likeability	3.74 (0.79)	4.16 (0.76)	0.07
Intelligence	3.91 (0.78)	3.96 (1.00)	0.73
Safety	3.63 (0.67)	3.83 (0.71)	0.13

The feedback modalities resulted in an observable increase in participants' perception of the agent's anthropomorphism (Mean = 3.06, $SD = 0.71$, $p = 0.01$) compared to the base condition (Mean = 2.58, $SD = 0.65$). There was also a noticeable increase in animacy (Mean = 3.58, $SD = 0.68$, $p = 0.01$) from the base condition (Mean = 3.09, $SD = 0.59$). While overall likeability improved (Mean = 4.16, $SD = 0.76$), it did not show statistical significance ($p = 0.07$) against the base condition (Mean = 3.74, $SD = 0.79$). No significant changes were noted in perceived intelligence (Mean = 3.96, $SD = 1.00$, $p = 0.73$) from the base condition (Mean = 3.91, $SD = 0.78$) or in perceived safety (Mean = 3.83, $SD = 0.71$, $p = 0.13$) compared to the base condition (Mean = 3.63, $SD = 0.67$).

In relation to participants' overall impression, openness and willingness to interact with the agent, displayed in Table 4, overall impression observed a meaningful increase in the feedback modalities condition (Mean = 4.00, $SD = 1.14$, $p = 0.05$) compared to the base condition (Mean = 3.61, $SD = 0.98$). As for openness to self-express, no significant change was seen for the feedback modalities condition (Mean = 3.83, $SD = 1.25$, $p = 0.33$) compared to the base condition (Mean = 3.67, $SD = 1.33$). Participants also reported higher willingness to interact again with the agent in the future for the feedback modalities condition (Mean = 4.06, $SD = 1.00$, $p = 0.04$) as opposed to the Base Condition (Mean = 3.61, $SD = 1.14$).

Table 4: Participant Ratings on Agent Impression, Openness to Self-Expression, and Willingness for Future Interactions

Factor	1: Base Condition	2: Modals Condition	p-value
Impression	3.61 (0.98)	4.00 (1.14)	0.05
Openness	3.67 (1.33)	3.83 (1.25)	0.33
Willingness	3.61 (1.14)	4.06 (1.00)	0.04

4.4 Effect of VR and Chatbot Experience on Agent Impression

Pearson correlation analysis was conducted to assess the relationships between participants' experience with VR and chatbots and their perceptions of the agent. No significant correlation was found between VR experience and perceived likeability ($r = -0.17$, $p = 0.495$), willingness to engage ($r = 0.14$, $p = 0.583$), or overall impression ($r = -0.06$, $p = 0.817$).

Contrastingly, chatbot experience yielded a significant positive correlation with perceived likeability ($r = 0.51$, $p = 0.030$), indicating a relationship between familiarity with chatbot interactions and positive reception of the agent. However, chatbot experience was not significantly correlated with willingness to engage ($r = 0.43$, $p = 0.078$) or overall impression ($r = 0.24$, $p = 0.337$).

Additionally, participants' attitudes, taken from the NARS questionnaire, showed a significant negative correlation with agent likeability ($r = -0.64$, $p = 0.005$), as well as overall impression ($r = -0.56$, $p = 0.016$). No significant correlation was observed between NARS scores and willingness to engage ($r = -0.31$, $p = 0.207$). The negative correlation with likeability underlines the influence of preconceived attitudes towards agents on user perceptions in interactive scenarios.

5 DISCUSSION

As shown in Table 5, the majority of participants preferred having the feedback modalities ($N=12$); contrastingly, several participants either felt mixed opinions ($N=3$) or preferred the baseline condition ($N=3$).

Table 5: User Preferences

Did you prefer the feedback modalities condition?	Count
Yes	12
No	3
Maybe	3

Nonverbal turn-taking behaviour was the most preferred feedback modality, having the highest number of participants who chose it as their first choice ($N=9$), followed by filled pauses ($N=5$) which was very closely tracked by the visual feedback indicator ($N=4$). A summary of the rankings is shown in Figure 6.

Hypotheses 1 and 2 were supported by our results. The integration of combined feedback modalities significantly enhanced participants' sense of presence and immersion. Additionally, it markedly

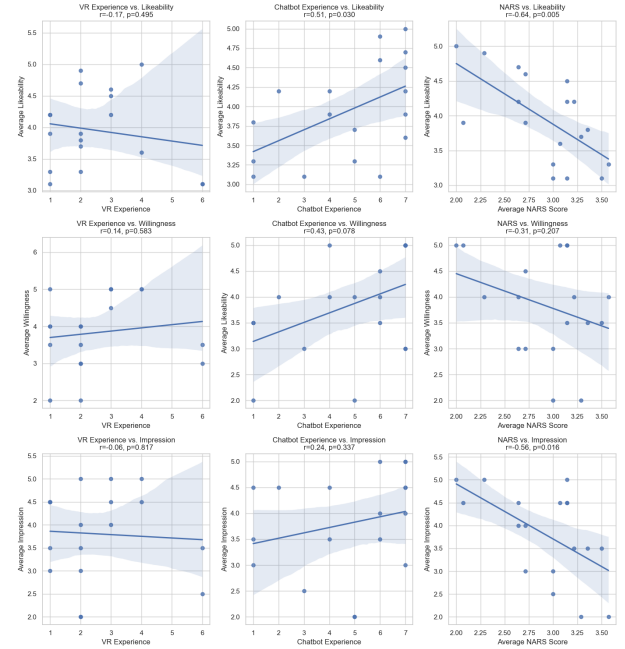


Figure 5: Correlations between individual's experiences with VR and chatbots (rated on a 7-point Likert scale; 1=never, 4=monthly, 7=almost daily) and NARS scores (1=strongly disagree, 7=strongly agree) in relation to agent likeability, willingness to interact, and overall impression (rated on a 5-point Likert scale; 1=strongly disagree, 5=strongly agree).

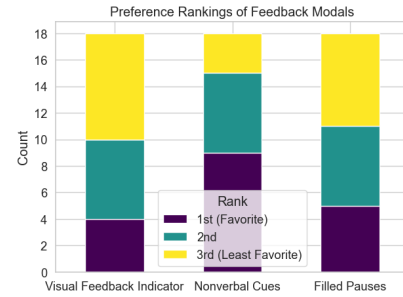


Figure 6: User rankings of the different feedback modalities

improved participants' perception of the agent's response time, consistent with previous findings [4, 19]. Employing the combined feedback modalities to signal understanding and processing of participants' queries led to participants evaluating the agent's response time more favourably as opposed to the condition without any feedback modalities present.

Hypotheses 3 and 4 were partially supported from our findings. We observed statistically significant results concerning participants' overall impression and willingness to interact with the agent again. However, similar to [4], there was no significant improvement in the agent's likeability. We also found a statistically significant positive correlation between increased chatbot experience and higher

agent likeability, supporting Hypothesis 4. This correlation was present for willingness to use the agent again, although no statistical significance was observed. Moreover, our investigation into the impact of VR experience on agent likeability and willingness to interact again yielded no statistically significant results.

Interview insights shed light on possible reasons for the small change in perceived likeability. Participants displayed diverse preferences and expectations in their interactions with the agent: some perceived the agent as a genuine social companion, more forgiving of flaws such as response delays, while others prioritised response efficiency over interaction quality. This divergence underscores the varied user expectations in agent interactions: while some appreciated the human-like engagement strategies, others remained focused on the functionality, seeking quicker responses irrespective of the interaction quality.

Participants' preferences among feedback modalities varied, with nonverbal turn-taking behaviours generally preferred, though no significant dominance over filled pauses or visual indicator feedback was observed. This could be attributed to nuances in execution. For instance, a poorly toned filled pause like "hm" can detract from the perceived intelligence of an agent, as it fails to convey appropriate emotional context, particularly in sensitive conversations (e.g., discussing depression). This emphasises the need for emotional intelligence in conversational agents, particularly when employing conversational expressions as fillers.

The effectiveness of visual feedback indicators was notably highlighted by participants who found them to reduce the awkwardness typically associated with delays in conversational response. Interestingly, some participants expressed a preference for minimal or no feedback modalities, citing experiences of mismatched filled pauses that led to perceptions of the agent as unnatural or less intelligent.

Additionally, Figure 7 illustrates the consequences of conversational mismatches on perceived agent intelligence, showing a decline in metrics for perceived intelligence and responsibility, while surprise increased, underscoring the importance of contextually appropriate responses.

Given these findings and our preliminary interpretations of the observed trends, we recommend several strategies to enhance user experience with LLM-powered agents in VR:

- (1) Employ nonverbal turn-taking behaviours more frequently and use filled pauses sparingly, ensuring they are contextually appropriate. This helps in maintaining the natural flow of interaction without imposing a robotic feel.
- (2) Tailor feedback modalities to individual preferences—use visual indicators for users less comfortable to socially interact with agents and filled pauses for those seeking richer dialogues.
- (3) Ensure the emotional tone of filled pauses matches the dialogue's context, particularly in sensitive discussions, to maintain user engagement and perceived empathy.
- (4) Incorporate additional non-verbal cues like facial expressions and eye movements to maintain a continuous sense of engagement and realism, thereby reducing the negative perception of latency in agent interactions.

With respect to adopting these ideas in other areas such as human-robot interaction, the visual feedback indicators may be

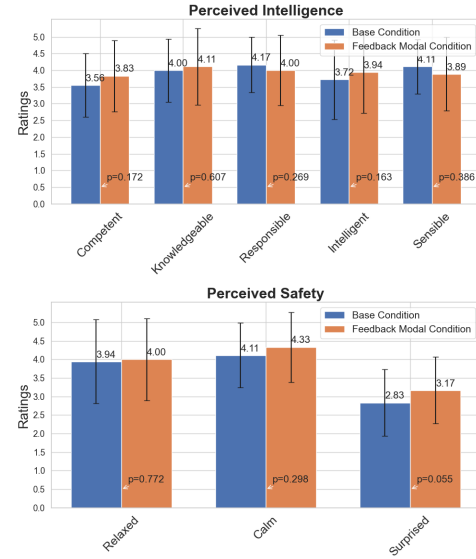


Figure 7: Godspeak Questionnaire results for individual sub-metrics of Perceived Intelligence and Safety

integrated in a diegetic manner (for example, differently coloured glowing strips of light on a part of the robot's body could indicate response state). Since physical embodiments like robots are likely to have a more limited number of gestures that can be made, it would be interesting to investigate how many gestures might be considered sufficient and to see if individuals may be more accepting towards these limitations.

6 CONCLUSION

Our study investigated the effects of multimodal feedback on perceived latency and user immersion in VR interactions with LLM-powered Embodied Conversational Agents. We found that filled pauses and nonverbal turn-taking behaviours, coupled with visual feedback indicators, significantly enhance users' sense of presence and immersion when interacting with LLM-powered agents within virtual environments. Notably, these feedback modalities improved perceived response time and user impressions of the agent, increasing their willingness for future interactions. While VR experience did not significantly correlate with agent impressions, familiarity with chatbots positively influenced agent likability, underscoring the importance of user background in interaction experiences. These results underscore the potential of multimodal feedback to mitigate negative perceptions of latency, enhancing the overall user experience in VR. Future research directions include: one, investigating the effect of each feedback modality separately; two, assessing the effect of the feedback modalities on sense of presence and immersion in interactions that are not mediated in VR; three, exploring the integration of facial expressions and gaze as additional feedback modalities to explore how they can further help to reduce perceived latency and enhance user immersion.

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